

18 Diffusion models

18.1 $\text{cov}[\bar{x}_t] = \mathbb{I}$

$$\begin{aligned}\text{cov}[\bar{x}_t] &= \text{cov}[\sqrt{1-\beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t] \\ &= \text{cov}[\sqrt{1-\beta_t} x_{t-1}] + \text{cov}[\sqrt{\beta_t} \epsilon_t] \\ &= (1-\beta_t) \underbrace{\text{cov}[\bar{x}_{t-1}]}_{=\mathbb{I}} + \beta_t \underbrace{\text{cov}[\epsilon_t]}_{=\mathbb{I}} \\ &= (1-\beta_t) \mathbb{I} + \beta_t \mathbb{I} = \mathbb{I} \quad \square\end{aligned}$$

18.2 $z = a \cdot \epsilon_1 + b \cdot \epsilon_2 \quad \epsilon_1, \epsilon_2 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \mathbb{I})$
 $E(z) = E(a \epsilon_1 + b \epsilon_2) = a E(\epsilon_1) + b E(\epsilon_2) = 0$
 $\text{Var}(z) = \text{Var}(a \epsilon_1 + b \epsilon_2) = a^2 \text{Var}(\epsilon_1) + b^2 \text{Var}(\epsilon_2) + 2 \text{cov}[\epsilon_1, \epsilon_2] = a^2 + b^2$

18.3 $z_3 = \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)} x + \sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)} \epsilon'$
 $\epsilon' \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \mathbb{I})$
 $z_3 = \sqrt{1-\beta_3} z_2 + \sqrt{\beta_3} \epsilon_3$
 $= \sqrt{1-\beta_3} [\sqrt{(1-\beta_2)(1-\beta_1)} x + \sqrt{1-(1-\beta_2)(1-\beta_1)} \epsilon] + \sqrt{\beta_3} \epsilon_3$
 $= \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)} x + \sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)} \epsilon + \sqrt{\beta_3} \epsilon_3$
 $\stackrel{\text{i.i.d.}}{=} \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)} x + \sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)} \epsilon' \quad \text{w.r. } \epsilon' = \epsilon + \epsilon_3$

18.4 $\text{hr } \text{Norm}_w(Aw, b) \propto \text{Norm}_w[(A^T B^{-1} A)^T A^T B^{-1} v, (A^T B^{-1} A)^{-1}]$

$$\begin{aligned}\text{Norm}_w(Aw, b) &\propto \exp[-0.5(v-Aw)^T B^{-1}(v-Aw)] \\ &= \exp[-0.5(v^T B^{-1} - (Aw)^T B^{-1})(v-Aw)] \\ &= \exp[-0.5(v^T B^{-1} v - v^T B^{-1} A w - w^T A^T B^{-1} v + w^T A^T B^{-1} A w)] \\ &\propto \exp[-0.5(-2v^T B^{-1} A w + w^T A^T B^{-1} A w)] \\ &\propto \exp[-0.5(v^T B^{-1} A (A^T B^{-1} A)^T A^T B^{-1} v - 2v^T B^{-1} A w \\ &\quad + w^T A^T B^{-1} A w)] \\ &= \exp[-0.5(w - (A^T B^{-1} A)^{-1} A^T B^{-1} v)^T (A^T B^{-1} A) \\ &\quad \cdot (w - (A^T B^{-1} A)^{-1} A^T B^{-1} v)] \\ &\propto \text{Norm}_w[(A^T B^{-1} A)^T A^T B^{-1} v, (A^T B^{-1} A)^{-1}] \quad \square\end{aligned}$$

18.5 $\text{hr } \text{Norm}_x(a, A) \text{Norm}_x(b, B) \propto \text{Norm}_x[(A^T + B^T)^T (A^T a + B^T b), (A^T + B^T)^{-1}]$

$$\begin{aligned}\text{Norm}_x(a, A) \text{Norm}_x(b, B) &\propto \exp[-0.5(x-a)^T A^{-1}(x-a) - (x-b)^T B^{-1}(x-b)] \\ &= \exp[-0.5(x^T (A^{-1} + B^{-1}) x - 2x(A^{-1} a + B^{-1} b) + a^T A^{-1} a + b^T B^{-1} b)] \\ &\propto \exp[-0.5(x^T (A^{-1} + B^{-1}) x - 2x(A^{-1} a + B^{-1} b))] \\ &\propto \exp[-0.5(x^T (A^{-1} + B^{-1}) x - 2x(A^{-1} a + B^{-1} b) + x^T (A^{-1} + B^{-1})^{-1} (A^{-1} a + B^{-1} b))] \\ &= \exp[-0.5(x - (A^{-1} + B^{-1})^{-1} (A^{-1} a + B^{-1} b))^T (A^{-1} + B^{-1}) (x - (A^{-1} + B^{-1})^{-1} (A^{-1} a + B^{-1} b))] \\ &\propto \text{Norm}_x[(A^{-1} + B^{-1})^{-1} (A^{-1} a + B^{-1} b), (A^{-1} + B^{-1})^{-1}] \quad \square\end{aligned}$$

18.6 $q(\bar{z}_{t-1} | z_t, x) \propto \text{Norm}_{\bar{z}_{t-1}}\left(\frac{1}{1-\beta_t} z_t, \frac{\beta_t}{1-\beta_t} \mathbb{I}\right) \text{Norm}_{\bar{z}_{t-1}}\left(\sqrt{\alpha_t} x, (1-\alpha_t) \mathbb{I}\right)$

Using 18.5: $q(\bar{z}_{t-1} | z_t, x) \propto \text{Norm}_{\bar{z}_{t-1}}[\mu, \Sigma]$

$$\begin{aligned}\text{with } \mu &= \frac{\beta_t(1-\alpha_t)}{1-\alpha_t} \left(\frac{1-\beta_t}{\beta_t(1-\beta_t)} z_t + \frac{\sqrt{\alpha_t}}{1-\alpha_t} x \right) \\ &= \frac{(1-\alpha_t)}{1-\alpha_t} \sqrt{1-\beta_t} z_t + \frac{\beta_t \sqrt{\alpha_t}}{1-\alpha_t} x\end{aligned}$$

$$\begin{aligned}\text{and } \Sigma &= \left(\frac{1-\beta_t}{\beta_t} + \frac{1}{1-\alpha_t} \right)^{-1} \mathbb{I} = \left(\frac{(1-\beta_t)(1-\alpha_t) + \beta_t}{\beta_t(1-\alpha_t)} \right)^{-1} \mathbb{I} \\ &= \left(\frac{1-\beta_t-\alpha_t+\beta_t\alpha_t+\beta_t}{\beta_t(1-\alpha_t)} \right)^{-1} \mathbb{I} \\ &= \left(\frac{1-\alpha_t+\beta_t\alpha_t}{\beta_t(1-\alpha_t)} \right)^{-1} \mathbb{I} \\ &= \left(\frac{1-(1-\beta_t)\alpha_t}{\beta_t(1-\alpha_t)} \right)^{-1} \mathbb{I} = \frac{\beta_t(1-\alpha_t)}{1-\alpha_t} \mathbb{I} \\ &\quad \text{w. } \alpha_t = (1-\beta_t)\alpha_t + 1 - 1\end{aligned}$$

18.7 $E[\log[\phi_1, \dots, \phi_T] | x] \approx \int q(\bar{z}_{1:T} | x) \left(\log[p(\bar{z}_{1:T}, \phi)] + \sum_{t=2}^T \log\left[\frac{p(\bar{z}_{t-1}, \phi_t)}{q(\bar{z}_{t-1}, z_t, x)}\right] \right) d\bar{z}_{1:T}$

$$= E\left[\log[p(\bar{z}_{1:T}, \phi)] - \sum_{t=2}^T E_{q(\bar{z}_{t-1})} \left[\log\left[\frac{q(\bar{z}_{t-1}, z_t, x)}{p(\bar{z}_{t-1}, z_t, \phi_t)}\right] \right] \right]$$

$$\begin{aligned}E_{q(\bar{z}_{t-1})} [\log[p(\bar{z}_{t-1}, \phi_t)]] &= \int q(\bar{z}_{t-1} | x) \log[p(\bar{z}_{t-1}, \phi_t)] d\bar{z}_{t-1} \\ &= \int q(\bar{z}_{1:T} | x) \log[p(\bar{z}_{t-1}, \phi_t)] d\bar{z}_{1:T}\end{aligned}$$

$$\begin{aligned}- \sum_{t=2}^T E_{q(\bar{z}_{t-1})} \left[\log\left[\frac{q(\bar{z}_{t-1}, z_t, x)}{p(\bar{z}_{t-1}, z_t, \phi_t)}\right] \right] &= - \sum_{t=2}^T \int q(\bar{z}_{t-1} | x) \log\left[\frac{q(\bar{z}_{t-1}, z_t, x)}{p(\bar{z}_{t-1}, z_t, \phi_t)}\right] d\bar{z}_{t-1} \\ &= - \sum_{t=2}^T \int q(\bar{z}_{t-1} | x) \int q(\bar{z}_{t-1} | z_t, x) \log\left[\frac{p(\bar{z}_{t-1}, z_t, \phi_t)}{q(\bar{z}_{t-1}, z_t, x)}\right] d\bar{z}_{t-1} dz_t \\ &= - \sum_{t=2}^T \int \int q(\bar{z}_{t-1}, z_t, x) q(\bar{z}_{t-1} | x) \log[\cdot] d\bar{z}_{t-1} dz_t \\ &= - \sum_{t=2}^T \int \int q(\bar{z}_{t-1}, z_t, x) \log[\cdot] d\bar{z}_{t-1} dz_t \\ &= - \sum_{t=2}^T \int q(\bar{z}_{1:T} | x) \log[\cdot] d\bar{z}_{1:T} \\ &= \int q(\bar{z}_{1:T} | x) \sum_{t=2}^T \log[\cdot] d\bar{z}_{1:T} \quad \square\end{aligned}$$

18.8 $D_{KL}[\text{Norm}_w(a, A) \| \text{Norm}_w(b, B)] = \frac{1}{2} \left(\text{tr}[B^{-1}A] - d + (a-b)^T B^{-1}(a-b) + \log\left[\frac{|B|}{|A|}\right] \right)$

$$Pr[\bar{z}_{t-1} | z_t, \phi_t] = \text{Norm}_{\bar{z}_{t-1}}[\underbrace{\mu}_{\beta_t := \frac{(1-\alpha_t)}{1-\alpha_t} z_t}, \underbrace{\Sigma}_{\beta_t := \frac{\beta_t(1-\alpha_t)}{1-\alpha_t} \mathbb{I}}]$$

$$q(\bar{z}_{t-1} | z_t, x) = \text{Norm}_{\bar{z}_{t-1}}\left[\underbrace{\frac{(1-\alpha_t)}{1-\alpha_t} \sqrt{1-\beta_t} z_t + \frac{\sqrt{\alpha_t} \beta_t}{1-\alpha_t} x}_{\mu}, \underbrace{\frac{\beta_t(1-\alpha_t)}{1-\alpha_t} \mathbb{I}}_{\Sigma}\right]$$

$$\begin{aligned}\Rightarrow D_{KL}[q(\bar{z}_{t-1} | z_t, x) \| Pr[\bar{z}_{t-1} | z_t, \phi_t]] &= \frac{1}{2} \left\| \left(\frac{(1-\alpha_t)}{1-\alpha_t} \sqrt{1-\beta_t} z_t + \frac{\sqrt{\alpha_t} \beta_t}{1-\alpha_t} x \right) - \mu(\bar{z}_{t-1}, \phi_t) \right\|^2 + \log\left[\frac{|B|}{|A|}\right]\end{aligned}$$

18.9 $\alpha_t = \prod_{s=1}^t 1-\beta_s$

$$\Rightarrow \sqrt{\frac{\alpha_t}{\alpha_{t-1}}} = \sqrt{1-\beta_t}$$

$$\sqrt{\frac{\alpha_t}{\alpha_{t-1}}} = \sqrt{\frac{\prod_{s=1}^t 1-\beta_s}{\prod_{s=1}^{t-1} 1-\beta_s}} = \sqrt{\frac{\prod_{s=1}^{t-1} (1-\beta_s) (1-\beta_t)}{\prod_{s=1}^{t-1} (1-\beta_s)}} = \sqrt{1-\beta_t} \quad \square$$

18.10 $\alpha_t = \prod_{s=1}^t 1-\beta_s$

$$\frac{(1-\alpha_{t-1})(1-\beta_t) + \beta_t}{(1-\alpha_t)(1-\beta_t)} = \frac{1}{\sqrt{1-\beta_t}}$$

$$\frac{(1-\alpha_{t-1})(1-\beta_t) + \beta_t}{(1-\alpha_t)(1-\beta_t)} = \frac{1-\alpha_t - \alpha_{t-1}(1-\beta_t) + \beta_t}{(1-\alpha_t)\sqrt{1-\beta_t}}$$

$$= \frac{1-\alpha_t(1-\beta_t)}{(1-\alpha_t)\sqrt{1-\beta_t}} = \frac{1-\alpha_{t-1}-\alpha_{t-1}\beta_t}{(1-\alpha_t)\sqrt{1-\beta_t}}$$

$$= \frac{(1-\alpha_t)}{(1-\alpha_t)\sqrt{1-\beta_t}} = \frac{1}{\sqrt{1-\beta_t}}$$

18.11

$$\frac{1}{2} \epsilon_t^2 \|x_i - \mu_i(\bar{z}_{t-1}, \phi_t)\|^2 = \frac{1}{2} \epsilon_t^2 \left\| \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} g_t(\bar{z}_{t-1}, \phi_t) - \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} \epsilon_{it} \right\|^2$$

$$\text{with } x_i = \frac{1}{\sqrt{\alpha_t}} z_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}} \epsilon_{it}$$

$$\mu_i(\bar{z}_{t-1}, \phi_t) = \frac{1}{\sqrt{1-\alpha_t}} z_t - \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} g_t(\bar{z}_{t-1}, \phi_t)$$

$$\frac{1}{2} \epsilon_t^2 \|x_i - \mu_i(\bar{z}_{t-1}, \phi_t)\|^2$$

$$= \frac{1}{2} \left\| \frac{1}{\sqrt{\alpha_t}} z_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}} \epsilon_{it} - \frac{1}{\sqrt{1-\alpha_t}} z_t + \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} g_t(\bar{z}_{t-1}, \phi_t) \right\|^2$$

$$\stackrel{\text{B.14.17}}{=} - \frac{\sqrt{1-\alpha_t}}{\sqrt{1-\alpha_t}} \epsilon_{it} + \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} g_t(\bar{z}_{t-1}, \phi_t)$$

$$= \frac{1-\alpha_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} \epsilon_{it}$$

$$= \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} g_t(\bar{z}_{t-1}, \phi_t) - \frac{\beta_t}{\sqrt{1-\alpha_t} \sqrt{1-\beta_t}} \epsilon_{it} \quad \square$$

18.12 > Transformer decodes: lower sampling temperature, top-k sampling, etc.

> GATs: Truncated prior

> LDU: Variance scaling in latent space

⇒ Not inevitable! Depends on goal: coarse → use all output
 high co-f. → restrict